

A Defect-Fiber Unified Framework for Higgs, Flavor, Neutrino Structure, and Cosmological Freeze-Out

中文参考标题：一种用于 Higgs、Flavor、Neutrino 结构及宇宙冷却冻结的缺陷—纤维统一框架

摘要 (Abstract)

We propose a defect-fiber unified framework in which Higgs structure, flavor hierarchies, neutrino suppression, and electroweak freeze-out are jointly organized as low-energy readouts of a fibered internal spectrum supported on cosmological defects. The framework is built from a temperature-dependent order-parameter field Φ , a defect worldvolume $\mathcal{D}$, and an internal one-dimensional fiber $\sigma \in [0, L]$ with non-equivalent boundary conditions. Within this setup, the lowest visible weak-doublet mode is naturally identified with the Higgs sector, while the next low-lying modes organize flavor structure. A unified bridge kernel generates mass matrices, mixing patterns, and CP structure across quark, charged-lepton, and neutral sectors. We further formulate a cosmological freeze-out chain $T_c \gtrsim T_{\text{form}} > T_{\text{split}} > T_H$, in which the Standard Model-like low-energy structure emerges as a stable residual readout after defect formation, fiber stabilization, internal splitting, and Higgs softening. A complete toy execution pipeline is presented, yielding coherent outputs for Higgs, electroweak masses, neutrino suppression, CKM, PMNS, and CP hierarchy. We also provide a quasi-no-go comparison showing that natural alternatives such as S^1 , $\mathbb{R}$, $\mathbb{R}_+$, or symmetric intervals fail to simultaneously support the key phenomenological features realized in the asymmetric finite-interval case. The present framework is not yet a full fundamental theory, but it defines a constrained and testable route toward a unified organization of Higgs, flavor, neutrino, and cosmological freeze-out physics.

1. Introduction

The Standard Model provides a remarkably successful low-energy description of particle physics, yet several of its most central structures remain largely input rather than output. These include the Higgs sector, flavor hierarchies, the pattern of quark and lepton mixing, the strong disparity between quark and leptonic CP structure, and the smallness of neutrino masses. In standard practice, these sectors are often treated separately: Higgs physics is encoded in one scalar sector, flavor is introduced through Yukawa couplings, neutrino suppression requires an additional mechanism, and cosmological history is usually discussed independently of their internal structural origin.

This separation motivates the search for a framework in which these sectors are not merely coexisting ingredients, but different low-energy readouts of a common higher organizing structure. The goal of the present work is not to claim a final theory of everything, nor to replace the Standard Model as an experimentally verified low-energy theory. Rather, the goal is more focused and more modest: to construct a constrained framework in which Higgs structure, flavor organization, neutrino suppression, and cosmological freeze-out can be described within a single defect-supported spectral geometry.

The central idea of this paper is that the low-energy particle sector may be organized by an internal one-dimensional fiber attached to cosmological defects generated by an order-parameter

field. The internal fiber supports a discrete spectrum. Its lowest visible mode is identified with the Higgs sector, while higher low-lying modes organize the flavor tower. A unified bridge kernel generates mass matrices, mixing patterns, and CP phases as projections on this same spectral background. In this picture, the Standard Model-like low-energy world is not fundamental from the outset; rather, it appears as a residual stable readout after the universe cools through a sequence of critical windows.

The framework developed below is therefore best viewed as a defect-fiber unification model for Higgs, flavor, neutrino structure, and cosmological freeze-out. It is constrained enough to be nontrivial, organized enough to generate toy observables, and explicit enough to admit quasi-no-go comparison with alternative internal topologies. At the same time, it is not yet a completed fundamental theory: gauge coupling unification is not fully derived, full quantum completion is absent, and some geometric choices remain physically rather than mathematically unique. These limitations are explicitly acknowledged and form part of the intended scope of the paper.

1. Minimal Geometric Setup

We consider a cosmological background endowed with a spacetime metric $g_{\mu\nu}$ and a temperature-dependent order-parameter field Φ . The central structural assumption is that the low-energy particle sector is not introduced directly in the bulk as a fully fundamental set of independent degrees of freedom, but emerges as a defect-supported spectral readout after the order parameter undergoes a cosmological transition.

The minimal object set is $(a(t), g_{\mu\nu}, \Phi, \mathcal{D}, \sigma, \mathbb{U}, \Psi, \mathcal{A})$.

Here $a(t)$ denotes the cosmological scale factor, Φ is the order-parameter field responsible for generating a defect-supporting phase, $\mathcal{D}$ is the resulting defect worldvolume, σ is an internal fiber coordinate supported on $\mathcal{D}$, $\mathbb{U}$ is an internal mother field controlling splitting structure, Ψ denotes matter modes, and $\mathcal{A}$ denotes gauge-sector connections.

The essential geometric claim of the framework is that the internal coordinate σ should not be interpreted as a generic free extra dimension, but as a defect-supported internal fiber coordinate. In the present formulation, the minimal viable fiber topology is taken to be a finite interval $F_\sigma \simeq I = [0, L]$, equipped with non-equivalent boundary conditions at the two ends.

This choice is motivated by phenomenological and dynamical requirements rather than by geometric aesthetics alone. The framework requires a geometry that simultaneously supports: 1. a discrete low-lying spectrum rather than a generic continuum, 2. a singled-out lowest visible mode for the Higgs sector, 3. a higher low-mode tower suitable for flavor organization, 4. near-neighbor dominant spectral mixing with suppressed long-range coupling, 5. asymmetric branch response across sectors, 6. compatibility with a finite defect width generated by Φ , 7. a temperature-dependent freezing of the effective internal size.

The interval geometry with asymmetric boundaries is the minimal candidate that naturally satisfies these requirements. Other obvious alternatives, such as S^1 , $\mathbb{R}$, $\mathbb{R}_+$, or a symmetric interval, will be discussed later and shown to fail at least one of the essential phenomenological conditions.

At this stage, the geometric interpretation is therefore:

σ is a defect-supported finite internal fiber coordinate, whose size and spectral structure are controlled by cosmological defect formation.} }
 This perspective will be central throughout the rest of the paper.

1. Unified Geometric Action

We now introduce the minimal unified action that organizes the framework. The total action is written schematically as $S_{\rm geom}$

$S_{\rm grav} + S_{\Phi} + S_{\mathbb{U}} + S_{\mathcal{A}} + S_{\Psi} + S_{\rm bridge}$.
 Each piece has a distinct structural role.

3.1 Gravitational sector

The gravitational part is taken minimally as $S_{\rm grav}$

$\int d^4x \sqrt{-g} \frac{M_P^2}{2} R$.
 At the present stage, gravity is not yet fully unified with all internal sectors at the level of a final microscopic principle. Instead, it supplies the cosmological background in which the order parameter Φ evolves and defects form. Nevertheless, Φ and the defect sector are not external to gravity; they are dynamical sources in spacetime and therefore couple back to geometry through their stress-energy.

3.2 Order-parameter sector

The order-parameter field is described by S_{Φ}

$\int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_{\mu} \Phi \partial_{\nu} \Phi - V_{\Phi}(\Phi, T) \right]$.

The temperature-dependent potential is chosen in minimal form as
 $V_{\Phi}(\Phi, T)$

$$\lambda_{\Phi}(\Phi^2 - v_{\Phi}^2)^2 + \frac{1}{2} c_T T^2 \Phi^2.$$

This potential has the following qualitative features.

When T is high enough, the effective quadratic term is positive and $\Phi=0$ is stabilized. In this regime, the system remains in a high-symmetry phase and no stable defect-supported internal fiber exists. When the universe cools below a critical temperature, the sign of the effective quadratic term changes, Φ becomes unstable around the origin, and the system can roll into nontrivial vacua $\Phi \sim \pm v_{\Phi}$, allowing kink/domain-wall-like defect solutions. These defects then support the internal spectral structure.

Thus, in the present framework, Φ plays a dual role: • it is the cosmological order parameter of a background transition, • it is also the generator of the finite internal geometry on which the low-energy particle sector is organized.

3.3 Internal mother field

The internal mother field $\mathbb{U}$ is defined on the defect worldvolume together with the internal fiber:
 $S_{\{\mathbb{U}\}}$

$$\int_{\mathcal{D}} d^p x \int d\sigma \sqrt{-h},$$

$$\left[|\mathbb{U}|^2 \right]$$

$$V_{\rm split}(\mathbb{U}).$$

The corresponding internal splitting potential is not assumed to provide a full microscopic derivation of the Standard Model, but it is designed to

stabilize a nontrivial residual structure.

A schematic minimal form is $V_{\rm split}(\mathbb{U})$

$$\alpha \big(|\mathbb{U}|^2 - \lambda v^2 \big)^2 + \beta \, \mathrm{Tr} \, C_{\rm eq}(\mathbb{U}) + \gamma \, \mathrm{Tr} \, C_{\rm lock}(\mathbb{U}).$$

Here: • the α -term controls the overall nonzero scale of the internal mother field, • the β -term penalizes unwanted equivalence directions and stabilizes internal differentiation, • the γ -term encodes locking between internal sectors.

This form is intentionally schematic, because what matters most at this stage is the existence of a controlled internal splitting mechanism rather than a unique microscopic completion.

3.4 Gauge and matter sectors

The gauge structure is represented by $S_{\mathcal{A}}$

$-\frac{1}{4} \int_{\mathcal{D}} d^p \xi \int d\sigma \sqrt{-h} \, F_{MN} F^{MN}$, and matter modes by S_{Ψ}

$$\int_{\mathcal{D}} d^p \xi \int d\sigma \sqrt{-h} \, \bar{\Psi}_i \Gamma^M D_M \Psi_i.$$

These are standard kinetic structures. The novelty here is not in their bare form, but in the fact that the relevant low-energy matter modes and their mixing data are to be understood as projections on a defect-supported internal spectrum rather than as arbitrary fundamental generations.

3.5 Bridge sector

The final piece is the bridge kernel: $S_{\rm bridge}$

$$\int_{\mathcal{D}} d^p \xi \int d\sigma \sqrt{-h} \, \bar{\Psi}_L \, \mathrm{Tr} \, K[\mathbb{U}, \sigma] \, \Psi_R + \text{h.c.}$$

This term is the origin of mass matrices, mixing structure, and CP texture. In the present framework, Yukawa-like structures are not taken as primitive unrelated constants; rather, they are low-energy projections of a common bridge operator defined on the internal fiber.

This is one of the main organizing principles of the entire construction.

1. Cosmological Defect Formation and Freeze-Out Chain

We now formulate the cosmological side of the framework. The main claim is that the low-energy Standard Model-like structure does not exist ab initio as a fixed set of microscopic objects. Instead, it appears only after the universe cools through a chain of transitions in which defects form, internal fibers stabilize, internal splitting becomes robust, and the weak lowest mode eventually softens.

4.1 Effective mass and critical temperature

From the temperature-dependent potential, $V_{\Phi}(\Phi, T)$

$\lambda_{\Phi}(\Phi^2 - v_{\Phi}^2)^2 + \frac{1}{2} c_T T^2 \Phi^2$, the effective mass around $\Phi=0$ is $m_{\text{eff}}^2(T) = -4\lambda_{\Phi} v_{\Phi}^2 + c_T T^2$.

The critical temperature T_c is defined by $m_{\text{eff}}^2(T_c) = 0$, hence $T_c = \frac{2}{\sqrt{c_T}} \sqrt{\lambda_{\Phi} v_{\Phi}^2}$.

For $T > T_c$, the origin is stable and no defect structure is available. For $T < T_c$, Φ becomes unstable at the origin and the system can roll toward a broken phase.

4.2 Defect formation

A sign change in the effective mass is not yet sufficient for fully formed defects. In an expanding universe, Hubble friction may delay or suppress actual domain formation. Therefore the natural defect-formation condition is $|m_{\text{eff}}(T)| \gtrsim H(T)$.

In the radiation-dominated regime, $H(T) \sim \frac{T^2}{M_{\text{Pl}}}$, so the defect-formation temperature T_{form} is determined by the relation $\sqrt{4\lambda_{\Phi} v_{\Phi}^2 - c_T T_{\text{form}}^2} \sim \frac{T_{\text{form}}^2}{M_{\text{Pl}}}$.

Since $T_c \ll M_{\text{Pl}}$ in the natural regime of interest, one typically obtains $T_{\text{form}} \lesssim T_c$, often extremely close to it from below. Thus the onset of instability and actual defect formation are nearby, but conceptually distinct.

4.3 Temperature-dependent fiber length

The defect width is controlled by the inverse effective mass scale, so the internal fiber size is taken to track the defect width: $L(T) = \frac{c_L}{|m_{\text{eff}}(T)|}$

$$\frac{c_L}{\sqrt{4\lambda_\Phi v_\Phi^2 - c_T T^2}}, \quad T < T_c.$$

This has a transparent interpretation.

Near T_c , the effective mass is small, so $L(T)$ is large: the internal fiber is long, diffuse, and not yet sharply spectral. As the universe cools, $|m_{\text{eff}}(T)|$ grows, the internal fiber contracts, and the low-lying spectral structure becomes progressively sharper. Eventually, at sufficiently low temperature, the internal scale effectively freezes.

Thus the low-energy particle structure is not simply present at all temperatures; it becomes legible only as the fiber geometry contracts and freezes.

4.4 Internal splitting and Higgs freeze-out

The internal 3-2-1-like residual structure requires not only a defect and a finite fiber, but also sufficiently strong internal locking. We therefore introduce a qualitative locking parameter $\Lambda_{\text{lock}}(T) \equiv \frac{E_{\text{lock}}}{T + \delta_\Phi^{-1}(T)}$, with $\delta_\Phi^{-1}(T) \sim |m_{\text{eff}}(T)|$.

The internal splitting stabilizes only when $\Lambda_{\text{lock}}(T) \gtrsim 1$. This defines a temperature T_{split} , below which the internal residual structure is robust enough to support the low-energy organization used later.

The Higgs sector is then treated as a further low-temperature event. The weak-sector lowest mode has an effective coefficient $m_H^2(T) = \lambda_0^{(w)}(T) + \Delta_{\text{split}}(T) + \Delta_{\text{lock}}(T)$. At a lower temperature T_H , this coefficient crosses zero: $m_H^2(T_H) = 0$. For $T < T_H$, the weak lowest mode softens and condenses.

4.5 Freeze-out sequence

The natural cosmological ordering is therefore $T_c \gtrsim T_{\text{form}} > T_{\text{split}} > T_H$.

This leads to the following interpretation: • Above T_c : no defect-supported internal fiber, no low-energy particle organization. • Just below T_c : defects begin to form, but the internal structure is still diffuse. • Below T_{split} : the internal splitting becomes stable, the low-mode tower becomes meaningful, and flavor structure begins to freeze. • Below T_H : the weak lowest mode condenses and electroweak symmetry breaking is realized. • At much lower temperatures: the Standard Model-like low-energy observables become frozen residual readouts of this cosmological sequence.

This is the sense in which the present framework treats the Standard Model-like world as a cosmological freeze-out structure, rather than an eternally fixed microscopic starting point.

1. Fiber Spectrum, Boundary Asymmetry, and Geometric Constraint

We now turn to the internal spectrum itself.

5.1 Spectral problem

The internal fiber modes are defined by the Sturm–Liouville problem $-\partial_\sigma^2 u_n(\sigma) + V_\sigma(\sigma) u_n(\sigma) = \lambda_n u_n(\sigma)$, $\sigma \in [0, L]$.

The most important point is not the exact detailed form of V_σ , but the fact that the spectrum is discrete, ordered, and supports a nontrivial lowest mode.

A convenient minimal potential family is $V_\sigma(\sigma) = V_0 + V_1 \cos\left(\frac{\pi \sigma}{L}\right) + V_2 \cos\left(\frac{2\pi \sigma}{L}\right)$.

This family is already sufficient to encode: • a dominant first harmonic, • a second harmonic correction, • and a natural mechanism for distinguishing near-neighbor and longer-range mode couplings.

5.2 Boundary asymmetry

The boundary conditions are taken to be $(\partial_{\sigma+m_L} u)|_{\sigma=0}=0$, $\quad (\partial_{\sigma-m_R} u)|_{\sigma=L}=0$.

The non-equivalence $m_L \neq m_R$ is not a decorative generalization. It is the key to: • nontrivial lowest-mode localization, • branch differentiation, • effective spectral seesaw-like asymmetry, • and the later quark/lepton stiffness contrast.

Symmetric boundaries allow a discrete tower, but they strongly weaken the phenomenological differentiation mechanisms required by the framework.

5.3 Why a finite interval?

At this point, one can already identify the physical reasons why the interval is preferred: 1. It provides a clean discrete spectral tower. 2. It supports a singled-out lowest mode without excessive periodic degeneracy. 3. It naturally admits low-order harmonic structures of the form $\cos(\pi\sigma/L)$ and $\cos(2\pi\sigma/L)$, which later generate near-neighbor dominant mixing and long-range suppression. 4. It is naturally compatible with a finite defect width and a temperature-dependent freeze-out scale $L(T)$. 5. It allows two genuinely distinct boundaries, which are crucial to the asymmetry structure of the model.

5.4 Quasi-no-go logic

The present work does not prove a full mathematical uniqueness theorem for the internal fiber. However, it does establish a strong quasi-no-go argument.

Natural alternatives fail as follows: • S^1 : excessive periodic symmetry, no true boundary asymmetry, weaker singled-out lowest mode, degraded branch differentiation. • $\mathbb{R}$: no finite freeze-out length, no natural discrete low tower without additional confining structure. • $\mathbb{R}_+$: only one boundary, insufficient asymmetry structure, degraded clean flavor tower. • symmetric interval: retains discrete spectrum, but weakens the quark/lepton branch differentiation and spectral seesaw-like asymmetry.

Thus, within the minimal phenomenological requirements of the framework, the asymmetric interval behaves not as an arbitrary choice, but as the minimal surviving geometry.

5.5 Spectral interpretation of Higgs and flavor

Once the interval structure is adopted, the low-lying modes can be given a simple interpretation: • the lowest visible weak-doublet mode is associated with the Higgs sector, • the next low-lying modes organize the flavor tower, • and their bridge-induced overlaps generate mass matrices and mixing hierarchies.

This is the point where geometric constraint begins to connect directly to phenomenology.

1. Residual Standard-Model-Like Structure from the Internal Mother Field

We now turn to the question of how the low-energy Standard-Model-like structure is organized within the present framework. The purpose of this section is not to claim a complete first-principles derivation of all gauge and matter structure, but rather to show that the defect-supported internal geometry admits a minimal residual organization naturally aligned with the observed low-energy pattern.

6.1 Minimal internal decomposition

The internal mother field $\mathbb{U}$ is assumed to stabilize into a minimal split structure that can be represented schematically as $\mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}$.

The interpretation is straightforward: • the three-dimensional complex block is the minimal carrier of a nontrivial threefold internal structure, • the two-dimensional complex block is the minimal carrier of doublet structure, • the one-dimensional complex phase block carries the residual common phase.

This is not yet a derivation of the Standard Model gauge group in a strict high-energy unification sense, but it is the minimal residual internal organization capable of reproducing the observed threefold/doublet/phase pattern at low energy.

6.2 Residual symmetry structure

At the level of the unreduced internal decomposition, the natural unitary symmetry is $U(3) \times U(2) \times U(1)$.

After quotienting out redundant overall phases, one arrives at a residual structure aligned with $SU(3) \times SU(2) \times U(1)$.

The present framework does not claim that this reduction is already a full microscopic derivation of the Standard Model gauge symmetry. Rather, the claim is more modest and more precise: the internal mother field naturally supports a residual organization with the correct minimal symmetry pattern needed for low-energy phenomenology.

6.3 Seed doublets and visible representation classes

A useful way to organize the visible low-energy matter sector is through two minimal seed doublets, $(1, 2, -1)$, $\simeq (3, 2, \frac{1}{3})$, which then close into singlet sectors. In this language, one may recover the standard six minimal low-energy representation classes as the visible residue of the internal splitting pattern.

What matters for the present framework is that these sectors are not inserted as entirely unrelated building blocks. They are understood as branches of a common internal geometry whose residual structure survives cosmological freeze-out.

6.4 Electric charge

The electric charge operator is read in the standard residual form $Q = T_3 + \frac{Y}{2}$.

In the present construction, this is not treated as an arbitrary postulate detached from the rest of the framework. Rather, it is the natural low-energy charge reading associated with the residual weak doublet structure together with the surviving phase block. In particular, once the Higgs sector is later identified with the lowest visible weak-doublet mode, the usual unbroken electromagnetic generator emerges as the residual generator preserving the condensate.

The principal message of this section is therefore not that the entire Standard Model has been rigorously derived from a unique microscopic principle, but that the defect-fiber framework already supports a minimal residual internal organization structurally aligned with the observed low-energy gauge/matter pattern.

1. Higgs as the Lowest Visible Weak Fiber Mode

One of the central claims of the present framework is that the Higgs sector should not be viewed as an unrelated primitive input field. Instead, it is naturally interpreted as the lowest visible weak-doublet mode supported on the defect fiber spectrum.

7.1 Higgs identification

Let $u_0^{(w)}(\sigma)$ denote the lowest visible weak-sector fiber mode. The effective low-energy Higgs field is identified schematically with this mode: $H \sim u_0^{(w)}$.

This identification is one of the principal structural advantages of the framework. It provides a concrete reason why a singled-out scalar doublet exists at low energy: it is not fundamental in isolation, but the first visible weak spectral readout of the defect-supported fiber.

7.2 Effective Higgs potential

Projecting the internal dynamics onto the lowest weak mode yields an effective potential of the form $V_{\text{eff}}(H)$

$$m_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2.$$

The quadratic coefficient is not assumed to be negative from the start. Instead, it is generated dynamically through the competition between: • the weak-sector lowest spectral scale, • the internal splitting background, • and the locking background.

Accordingly, one writes $m_H^2(T) = \lambda_0^{(w)}(T) + \Delta_{\text{split}}(T) + \Delta_{\text{lock}}(T)$.

The Higgs softening temperature T_H is then defined by $m_H^2(T_H) = 0$.

For $T < T_H$, the coefficient becomes negative and the weak lowest mode condenses. The electroweak sector is thus not fundamental at arbitrarily high temperature; rather, it emerges as a late-stage low-energy effect after defect formation and internal splitting have already stabilized.

7.3 Electroweak breaking

Once the Higgs mode condenses, one obtains the usual electroweak pattern. With $\langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_H \end{pmatrix}$, the gauge boson masses take the standard residual form $m_W^2 = \frac{g^2 v_H^2}{4}$, $m_Z^2 = \frac{(g^2 + g'^2) v_H^2}{4}$, $m_\gamma = 0$.

Thus, the present framework preserves the familiar low-energy electroweak output while relocating its structural origin from a primitive Higgs scalar to the lowest visible weak fiber mode.

7.4 Physical interpretation

The crucial conceptual point is that Higgs condensation occurs after several earlier layers have already taken place: 1. cosmological instability of Φ , 2. defect formation, 3. fiber freeze-out, 4. internal 3-2-1 stabilization, 5. weak-mode softening.

This ordering is one of the framework's main claims. The Higgs sector is therefore not the first actor in the construction; it is a late emergent low-energy event within a deeper defect-supported spectral organization.

1. Unified Flavor and Neutrino Mechanism

We now turn to the part of the framework that is arguably its most distinctive feature: the joint organization of quark flavor, lepton flavor, neutrino suppression, and CP structure through one and the same defect-fiber spectral background.

The bridge kernel is taken in the minimal family $\mathcal{K}_R(\sigma)$

$$K_*(1+\rho \quad \Xi_R) \quad \left[1+\frac{\mu}{S_R} \cos \frac{\pi \sigma}{L} - \varepsilon_R \cos \frac{2\pi \sigma}{L} \right] e^{i\Theta_R(\sigma)}.$$

The ingredients have the following roles: • K_* : overall bridge scale, • Ξ_R : representation-complexity weight, • S_R : branch stiffness/softness factor, • μ : dominant near-neighbor spectral coupling scale, • ε_R : second-harmonic suppression, especially relevant to the 13 sector, • $\Theta_R(\sigma)$: phase texture, responsible for CP structure.

The corresponding mass matrix in sector R is $M_R^{\{nm\}}$

$$\int_0^L u_n(\sigma) \mathcal{K}_R(\sigma) u_m(\sigma) d\sigma.$$

This unified construction applies across: • quark sectors, • charged-lepton sectors, • neutral sectors.

8.2 Near-neighbor dominance

A major success of the framework is that the same interval geometry that was selected by the quasi-no-go analysis also naturally yields the low-order harmonic selection rules responsible for $|M_{12}|, |M_{23}| > |M_{13}|$.

This is not imposed as an abstract texture ansatz. It follows from the combination of: • an interval fiber spectrum, • a first harmonic bridge term, • a second harmonic correction.

Thus the framework gives a geometric explanation of near-neighbor dominance rather than merely postulating it.

8.3 Branch stiffness and flavor differentiation

The stiffness factor S_R plays a central phenomenological role. In the quark sector, the response to the same bridge perturbation is stiffer; in the charged-lepton sector it is softer; and in the neutral sector softer still. This allows the same mother kernel to generate small quark mixing and large lepton mixing without introducing entirely disconnected flavor models.

This point is essential. The framework does not require one flavor theory for quarks and another for leptons. Instead, it proposes that the same defect-fiber geometry is read differently by sectors of different stiffness.

8.4 Spectral seesaw-like suppression

The neutral sector exhibits an additional suppression mechanism. Since the lightest neutral modes and heavier neutral modes both live on the same spectral tower, their bridge coupling can yield a suppression of the form $m_{\nu, \text{light}} \sim \frac{m_D^2}{m_{\text{heavy}}}$, without requiring the framework to begin with a completely separate neutrino theory. This is not identical to conventional Majorana seesaw constructions, but it plays an analogous structural role at the level of the defect-supported internal spectrum.

8.5 CKM, PMNS, and CP

The matrices M_u, M_d, M_e, M_ν are diagonalized in the usual way, leading to $V_{\text{CKM}} = U_u^\dagger U_d, U_{\text{PMNS}} = U_e^\dagger U_\nu$.

The current toy execution chain shows that the framework can coherently realize: • small CKM angles, • large PMNS angles, • and a large separation between leptonic and quark CP measures, $|J_{\text{PMNS}}| \gg |J_{\text{CKM}}|$.

The significance of this result is not yet final numerical precision, but the fact that all these structures emerge from one and the same spectral background.

1. Toy Execution Pipeline and Global Toy Parameter Table

The framework is intended to be executable, not merely symbolic. A minimal toy implementation is organized through the chain $(\lambda_\Phi, v_\Phi, c_L, m_L, m_R, V_0, V_1, V_2, K^*, \rho, \mu, \theta) \rightarrow (\Phi, L, u_n, \lambda_n) \rightarrow (V_{\text{eff}}(H), m_h, m_W, m_Z) \rightarrow (M_u, M_d, M_e, M_\nu) \rightarrow (V_{\text{CKM}}, U_{\text{PMNS}}, J)$.

A benchmark toy execution produces the following qualitative outputs: • a kink/domain-wall background profile, • a finite frozen interval length, • a low-lying discrete spectrum, • a softened Higgs mode, • nonzero W and Z masses with $m_\gamma=0$, • a strongly suppressed neutral light scale, • a coherent quark/lepton flavor split, • refined small CKM structure, • refined large PMNS structure, • and a strong CP separation between lepton and quark sectors.

The importance of this section is methodological. The framework does not merely reinterpret known quantities after the fact; it defines an actual computational route from cosmological defect data to low-energy observables.

In this sense, the framework should be regarded as an executable unified toy model rather than a purely formal speculation.

1. Discussion, Boundary of Validity, and Falsifiability

The present framework is ambitious, but its status must be described with care.

10.1 What is already strong

Several aspects of the framework are already nontrivial and deserve emphasis: 1. It provides a common organizing background for Higgs, flavor, neutrino structure, and cosmological freeze-out. 2. It embeds the Higgs sector into the same spectral geometry that organizes flavor. 3. It ties low-energy observables to a cosmological sequence rather than treating them as timeless isolated inputs. 4. It constrains the internal fiber through quasi-no-go phenomenological logic. 5. It is executable as a toy pipeline and not merely conceptual.

10.2 What remains incomplete

At the same time, several major limitations remain: 1. gauge couplings are not yet fully derived from first principles; 2. the framework does not yet provide a full quantum completion; 3. the absolute neutrino scale still requires further suppression in realistic units; 4. the fiber quasi-uniqueness result is physically strong but not a strict mathematical classification theorem; 5. the toy benchmark is structurally coherent but not yet a precision fit to data.

These are not minor caveats. They define the current boundary between a strong unification candidate and a completed fundamental theory.

10.3 Falsifiability

The framework is falsifiable. Its most natural experimental vulnerabilities lie not necessarily in direct new-particle claims, but in correlated structures among existing observables.

Examples include: • correlated behavior among leptonic mixing angles and leptonic CP structure, • a relation between neutral-sector softness and neutrino lightness, • weak but nontrivial correlations between Higgs soft-mode data and flavor organization.

If future precision data establish that these sectors are completely independent, the framework would lose one of its central organizing claims.

Likewise, if interval-based asymmetric fiber structures are shown unable to reproduce stable large lepton/small quark separation while preserving Higgs low-mode behavior, the framework would be strongly disfavored.

10.4 Final position

The present work should therefore be viewed as a constrained defect-fiber unification framework rather than a completed theory of everything. It goes significantly beyond disconnected phenomenological ansätze by embedding multiple low-energy sectors into a common cosmological and spectral geometry, but it does not yet solve all ultraviolet, quantum, or coupling-unification questions.

Its strongest current value lies in demonstrating that Higgs, flavor, neutrino structure, and cosmological freeze-out can be organized coherently within a single defect-supported asymmetric fiber spectrum.